

Almost Optimal Inapproximability of Multidimensional Packing Problems

Sai Sandeep

Computer Science Department

Carnegie Mellon University

Pittsburgh, PA 15213

Email: spallerl@andrew.cmu.edu

Abstract—Multidimensional packing problems generalize the classical packing problems such as Bin Packing, Multiprocessor Scheduling by allowing the jobs to be d -dimensional vectors. While the approximability of the scalar problems is well understood, there has been a significant gap between the approximation algorithms and the hardness results for the multidimensional variants. In this paper, we close this gap by giving almost tight hardness results for these problems.

- 1) We show that Vector Bin Packing has no $\Omega(\log d)$ factor asymptotic approximation algorithm when d is a large constant, assuming $P \neq NP$. This matches the $\ln d + O(1)$ factor approximation algorithms (Chekuri, Khanna SICOMP 2004, Bansal, Caprara, Sviridenko SICOMP 2009, Bansal, Eliáš, Khan SODA 2016) upto constants.
- 2) We show that Vector Scheduling has no polynomial time algorithm with an approximation ratio of $\Omega((\log d)^{1-\epsilon})$ when d is part of the input, assuming $NP \not\subseteq ZPTIME(n^{(\log n)^{O(1)}})$. This almost matches the $O\left(\frac{\log d}{\log \log d}\right)$ factor algorithms (Harris, Srinivasan JACM 2019, Im, Kell, Kulkarni, Panigrahi SICOMP 2019). We also show that the problem is NP-hard to approximate within $(\log \log d)^{\omega(1)}$.
- 3) We show that Vector Bin Covering is NP-hard to approximate within $\Omega\left(\frac{\log d}{\log \log d}\right)$ when d is part of the input, almost matching the $O(\log d)$ factor algorithm (Alon *et al.*, Algorithmica 1998).

Previously, no hardness results that grow with d were known for Vector Scheduling and Vector Bin Covering when d is part of the input and for Vector Bin Packing when d is a fixed constant.

I. INTRODUCTION

Bin Packing and Multiprocessor Scheduling (also known as Makespan Minimization) are some of the most fundamental problems in Combinatorial Optimization. They have been studied intensely from the early days of approximation algorithms and have had a great impact on the field. These two are packing problems where we have n jobs with certain sizes, and the objective is to pack them into bins efficiently. In Bin Packing, each bin has unit size and the objective is to minimize the number of bins, while in Multiprocessor Scheduling, we are given a fixed number of bins and the objective is to minimize the maximum load in a bin. A relatively less studied but natural problem is Bin Covering, where the objective is to assign the jobs to the maximum number of bins such that in each bin, the load is at least

1. These problems are well understood in terms of approximation algorithms: all the three problems are NP-hard, and all of them have a Polynomial Time Approximation Scheme (PTAS) [dV81], [HS87], [CJK01].

In this paper, we study the approximability of the multidimensional generalizations of these three problems. The three corresponding problems are Vector Bin Packing, Vector Scheduling, and Vector Bin Covering. Apart from their theoretical importance, these problems are widely applicable in practice [Spi94], [ST12], [PTUW11] where the jobs often have multiple dimensions such as CPU, Hard disk, memory, etc.

In the Vector Bin Packing problem, the input is a set of n vectors in $[0, 1]^d$ and the goal is to partition the vectors into the minimum number of parts such that in each part, the sum of vectors is at most 1 in every coordinate. The problem behaves differently from Bin Packing even when $d = 2$: Woeginger [Woe97] proved¹ that there is no asymptotic² PTAS for 2-dimensional Vector Bin Packing, assuming $P \neq NP$. On the algorithmic front, the PTAS for Bin Packing [dV81] easily implies a $d + \epsilon$ approximation for Vector Bin Packing. When d is part of the input, this is almost tight: there is a lower bound of $d^{1-\epsilon}$ shown by [CK04]³. When d is a fixed constant⁴, much better algorithms are known [CK04], [BCS09], [BEK16] that get $\ln d + O(1)$ approximation guarantee. However, the best hardness factor (for arbitrary constant d) is still the APX-hardness result of the 2-dimensional problem due to Woeginger from 1997.

Closing this gap, either by obtaining a $O(1)$ factor algorithm or showing a hardness factor that is a function of d , has remained a challenging open problem. It is one of the ten open problems in a recent survey on multidimensional

¹Very recently, Ray [Ray21] found an oversight in Woeginger's original proof and gave a revised APX hardness proof for the problem.

²The asymptotic approximation ratio (formally defined in Section II) of an algorithm is the ratio of its cost and the optimal cost when the optimal cost is large enough. All the approximation factors mentioned in this paper for Vector Bin Packing are asymptotic.

³[CK04] actually give $d^{\frac{1}{2}-\epsilon}$ hardness, but it has been shown later (see e.g., [CKPT17]) that a slight modification of their reduction gives $d^{1-\epsilon}$ hardness.

⁴The algorithms are now allowed to run in time $n^{f(d)}$, for some function f .

scheduling problems [CKPT17]. It also appeared in a recent report by Bansal [Ban17] on open problems in scheduling. In fact, to the best of our knowledge, super constant integrality gap instances for the configuration LP relaxation of the problem were also not known. For the integer instances i.e. when the vectors are from $\{0, 1\}^d$ (which are the hard instances for Vector Scheduling and Vector Bin Covering), there is an asymptotic PTAS since there are $O_d(1)$ item types.

In the Vector Scheduling problem, given a set of n vector jobs in $[0, 1]^d$, and m identical machines, the objective is to assign the jobs to machines to minimize the maximum ℓ_∞ norm of the load on the machines. Chekuri and Khanna [CK04] introduced the problem as a natural generalization of Multiprocessor Scheduling and obtained a PTAS for the problem when d is a fixed constant. When d is part of the input, they obtained a $O(\log^2 d)$ factor approximation algorithm. They also showed that it is NP-hard to obtain a C factor approximation algorithm for the problem, for any constant C . Meyerson, Roytman, and Tagiku [MRT13] gave an improved $O(\log d)$ factor algorithm while the current best factor is $O\left(\frac{\log d}{\log \log d}\right)$ due to Harris and Srinivasan [HS19] and Im, Kell, Kulkarni, and Panigrahi [IKKP19]. The algorithm of Harris and Srinivasan [HS19] works for the more general setting of unrelated machines where each job can have a different vector load for each machine. However, no super constant hardness is known even in this unrelated machines setting.

In the Vector Bin Covering problem, the input is a set of n vectors in $[0, 1]^d$. The objective is to partition these into the maximum number of parts such that in each part, the sum of vectors is at least 1 in every coordinate. It is also referred to as “dual Vector Packing” in the literature. This problem is introduced by Alon *et al.* [AAC⁺98] who gave a $O(\log d)$ factor approximation algorithm. They also gave a d factor algorithm using a method from the area of compact vector approximation that outperforms the above algorithm for small values of d . On the hardness front, Ray [Ray21] showed that the 2-dimensional Vector Bin Covering problem is hard to approximate within a factor of $\frac{998}{997}$.

A. Our Results

We prove almost optimal hardness results for the three multidimensional problems discussed above.

1) *Vector Bin Packing*: For the Vector Bin Packing problem, we prove a $\Omega(\log d)$ asymptotic hardness of approximation when d is a large constant, matching the $\ln d + O(1)$ approximation algorithms [CK04], [BCS09], [BEK16], up to constants.

Theorem I.1. *There exists an integer d_0 and a constant $c > 0$ such that for all constants $d \geq d_0$, d -dimensional Vector Bin Packing has no asymptotic $c \log d$ factor polynomial time approximation algorithm unless $P = NP$.*

We obtain our hardness result via a reduction from the set cover problem on certain structured instances. In the set cover problem, we are given a set system $\mathcal{S} \subseteq 2^V$ on a universe V , and the goal is to pick the minimum number of sets from $\mathcal{S}$ whose union is V . Observe that Vector Bin Packing is a special case of the set cover problem with the vectors being the elements and every maximal set of vectors whose sum is at most 1 in every coordinate (known as “configurations”) being the sets. In fact, in the elegant Round & Approx framework [BCS09], [BEK16], the Vector Bin Packing problem is viewed as a set cover instance, and the algorithms proceed by rounding the standard set cover LP. Towards proving the hardness of Vector Bin Packing, we ask the converse: *Which families of set cover instances can be cast as d -dimensional Vector Bin Packing?*

We formalize this question using the notion of *packing dimension* of a set system $\mathcal{S}$ on a universe V : it is the smallest integer d such that there is an embedding $f : V \rightarrow [0, 1]^d$ such that a set $S \subseteq V$ is in $\mathcal{S}$ if and only if

$$\left\| \sum_{v \in S} f(v) \right\|_\infty \leq 1$$

If a set system has packing dimension d , then the corresponding set cover problem can be embedded as a d -dimensional Vector Bin Packing instance. However, it is not clear if the hard instances of the set cover problem have a low packing dimension. Indeed the instances in the $(1 - \epsilon) \ln n$ set cover hardness [Fei98] have a large packing dimension that grows with n , which we cannot afford as we are operating in the constant d regime. We get around this by starting our reduction from highly structured yet hard instances of set cover. In particular, we study simple bounded set systems which satisfy the following three properties:

- 1) The set system is *simple*⁵ i.e., every pair of sets intersect in at most one element.
- 2) The cardinality of each set is at most k , a fixed constant.
- 3) Each element in the family is present in at most $\Delta = k^{O(1)}$ sets.

Kumar, Arya, and Ramesh [KAR00] proved that simple set cover i.e., set cover with the restriction that every pair of sets intersect in at most one element, is hard to approximate within $\Omega(\log n)$. We observe that by modifying the parameters slightly in their proof, we can obtain the $\Omega(\log k)$ hardness of simple bounded set cover.

We prove that simple bounded set systems have packing dimension at most $k^{O(1)}$. Thus, the $\Omega(\log k)$ simple bounded set cover hardness translates to $\Omega(\log d)$ hardness of Vector Bin Packing when d is a constant. Note that the optimal value of the set cover instances can be made arbitrarily large in terms of k by starting with a Label Cover instance

⁵Simple set families are also known as linear set families.

with an arbitrarily large number of edges. Thus, our Vector Bin Packing hardness holds for asymptotic approximation as well.

Our upper bound on the packing dimension is obtained in two steps: First, we write the given simple bounded set system as an intersection of $(k\Delta)^{O(1)}$ structured simple bounded set systems on the same universe, and then we give an embedding using $(k\Delta)^{O(1)}$ dimensions bounding the packing dimension of these structured simple bounded set systems. This idea of decomposition into structured instances is inspired from a work of Chandran, Francis, and Sivadasan [CFS08] where an upper bound on the Boxicity of a graph is obtained in terms of its maximum degree. We believe that the packing dimension of set systems is worth studying on its own, especially in light of its close connections to the notions of dimension of graphs such as Boxicity.

2) *Vector Scheduling*: For the Vector Scheduling problem, we obtain a $\Omega((\log d)^{1-\epsilon})$ hardness under $\text{NP} \not\subseteq \text{ZPTIME}(n^{(\log n)^{O(1)}})$, almost matching the $O(\frac{\log d}{\log \log d})$ algorithms [HS19], [IKKP19].

Theorem 1.2. *For every constant $\epsilon > 0$, assuming $\text{NP} \not\subseteq \text{ZPTIME}(n^{(\log n)^{O(1)}})$, d -dimensional Vector Scheduling has no polynomial time $\Omega((\log d)^{1-\epsilon})$ -factor approximation algorithm when d is part of the input.*

We obtain the hardness result via a reduction from the Monochromatic Clique problem. In the Monochromatic-Clique(k, B) problem, given a graph $G = (V, E)$ with $|V| = n$ and parameters $k(n)$ and $B(n)$, the goal is to distinguish between the case when G is k -colorable and the case when in any assignment of k -colors to vertices of G , there is a clique of size B all of whose vertices are assigned the same color. When $B = 2$, this is the standard graph coloring problem. Note that the problem gets easier as B increases. Indeed, when $B > \sqrt{n}$, we can solve the problem in polynomial time by computing the Lovász theta function of the complement graph. We are interested in proving the hardness of the problem for B as large a function of n as possible, for some k . For example, given a graph that is promised to be k colorable, can we prove the hardness of assigning k colors to the vertices of the graph in polynomial time where each color class has maximum clique at most $B = \log n$?

The Monochromatic Clique problem was defined formally by Im, Kell, Kulkarni, and Panigrahi [IKKP19] in the context of proving lower bounds for online Vector Scheduling. It was also used implicitly in the $\omega(1)$ NP-hardness of Vector Scheduling by Chekuri and Khanna [CK04]. They proved (implicitly) that Monochromatic Clique is NP-hard when B is an arbitrary constant using a reduction from $n^{1-\epsilon}$ hardness of graph coloring. We observe that the same reduction combined with better hardness of graph chromatic

number [Kho01] proves the hardness of Monochromatic Clique when $B = (\log n)^\gamma$, for some constant $\gamma > 0$ under the assumption that $\text{NP} \not\subseteq \text{ZPTIME}(n^{(\log n)^{O(1)}})$.

We then amplify this hardness to $B = (\log n)^C$, for every constant $C > 0$. Our main idea in this amplification procedure is the notion of a stronger form of Monochromatic Clique where given a graph and parameters k, B, C , the goal is to distinguish between the case that G is k colorable vs. in any k^C coloring of G , there is a monochromatic clique of size B . It turns out that the graph coloring hardness of Khot [Kho01] already proves the hardness of this stronger variant of Monochromatic clique when $B = (\log n)^\gamma$ for any constant C . We then use lexicographic product of graphs to amplify this result into the hardness of original Monochromatic Clique problem with $B = (\log n)^C$ for any constant C under the same assumption that $\text{NP} \not\subseteq \text{ZPTIME}(n^{(\log n)^{O(1)}})$. This directly gives the required hardness of Vector Scheduling using the reduction in [CK04].

The Vector Scheduling problem is also closely related to the Balanced Hypergraph Coloring problem where the input is a hypergraph H and a parameter k , and the objective is to color the vertices of H using k colors to minimize the maximum number of times a color appears in an edge. We use this connection to improve upon the NP-hardness of the problem:

Theorem 1.3. *For every constant $C > 0$, d -dimensional Vector Scheduling is NP-hard to approximate within $\Omega((\log \log d)^C)$ when d is part of the input.*

Consider the case when each vector job is from $\{0, 1\}^d$. In this setting, we can view each coordinate as an edge in a hypergraph, and each vector corresponds to a vertex of the hypergraph. The goal is to find a m -coloring of vertices of the hypergraph i.e., an assignment of the vectors to m machines to minimize the maximum number of monochromatic vertices in an edge, which directly corresponds to the maximum load on a machine.

This problem of coloring a hypergraph to ensure that no color appears too many times in each edge is known as Balanced Hypergraph Coloring. Guruswami and Lee [GL18] obtained strong hardness results for this problem when k , the uniformity of the hypergraph is a constant, using the Label Cover Long Code framework combined with analytical techniques such as the invariance principle. However, when k is super constant, the invariance principle based methods give weak bounds as the soundness of the Label Cover has to be at least exponentially small in k . Recently, using combinatorial tools to analyze the gadgets instead of the standard analytical techniques, improvements have been obtained for various hypergraph coloring problems [Bha18], [ABP19] in the super-constant inapproximability regime. We follow the same route and use combinatorial tools to

analyze the gadgets in the Label Cover Long Code framework and obtain better hardness of Balanced Hypergraph Coloring in the regime of super-constant uniformity k . The key combinatorial lemma used in our analysis was proved recently by Austrin, Bhangale, and Potukuchi [ABP20] using a generalization of the Borsuk-Ulam theorem.

The NP-hardness of Vector Scheduling follows directly from the hardness of Balanced Hypergraph Coloring using the above-described reduction. This NP-hardness result uses near-linear size Label Cover hardness results [MR10], [DS14]. By using the standard Label Cover hardness obtained by combining PCP Theorem and Parallel Repetition in the same reduction, we also prove an intermediate result bridging the above two hardness results for Vector Scheduling.

Theorem I.4. *There exists a constant $\gamma > 0$ such that assuming $NP \not\subseteq DTIME(n^{O(\log \log n)})$, d -dimensional Vector Scheduling is hard to approximate within $\Omega((\log d)^\gamma)$ when d is part of the input.*

3) *Vector Bin Covering:* For the Vector Bin Covering problem, we show $\Omega\left(\frac{\log d}{\log \log d}\right)$ hardness, almost matching the $O(\log d)$ factor algorithm [AAC+98].

Theorem I.5. *d -dimensional Vector Bin Covering is NP-hard to approximate within $\Omega\left(\frac{\log d}{\log \log d}\right)$ factor when d is part of the input.*

Similar to Vector Scheduling, the hard instances for the Vector Bin Covering are when each vector is in $\{0, 1\}^d$. Using the same connection as before, we view this problem as a hypergraph coloring problem where each edge of the hypergraph corresponds to a coordinate, and each vertex corresponds to a vector. Assigning the vectors to the bins such that in each bin, the sum is at least 1 in every coordinate corresponds to assigning colors to vertices of the hypergraph such that in every edge, all the colors appear. Such a coloring of the hypergraph with k colors where all the k colors appear in every edge is known as a k -rainbow coloring of the hypergraph.

Strong hardness results are known for approximate rainbow coloring [GL18], [ABP20], [GS20] when k is a constant. While these results give decent bounds in the super constant regime, they proceed via the hardness of Label Cover whose soundness is an inverse polynomial function of k . Because of this, in the NP-hard instances, the number of edges is at least doubly exponential in k . Instead, by losing a factor of 2 in the hardness, we give a reduction to the approximate rainbow coloring problem from Label Cover with no gap i.e., just a “Label Coverized” 3-SAT instance. In these hard instances, the number of edges is single exponential in k , proving that it is NP-hard to distinguish the case that a hypergraph with m edges has a rainbow coloring with $\Omega\left(\frac{\log m}{\log \log m}\right)$ colors vs. it cannot be rain-

bow colored with 2 colors⁶. This hardness of approximate rainbow coloring gives the required Vector Bin Covering hardness immediately using the earlier mentioned analogy between rainbow coloring and Vector Bin Covering.

B. Related Work

Online Algorithms. Multidimensional packing problems have been extensively studied in the online setting. For the d -dimensional Vector Bin Packing, the classical First-Fit algorithm [GGJY76] gives $O(d)$ competitive ratio, and Azar, Cohen, Kamara, and Shepherd [ACKS13] recently gave an almost matching $\Omega(d^{1-\epsilon})$ lower bound. For the d -dimensional Vector Scheduling, Im, Kell, Kulkarni, and Panigrahi [IKKP19] gave a $O\left(\frac{\log d}{\log \log d}\right)$ competitive online algorithm and proved a matching lower bound. For the d -dimensional Vector Bin Covering problem, Alon et al. [AAC+98] gave a $2d$ competitive algorithm and proved a lower bound of $d + \frac{1}{2}$.

Geometric variants. There are various natural geometric variants of Vector Bin Packing that have been studied in the literature. A classical problem of this sort is the 2-dimensional Geometric Bin Packing, where the input is a set of rectangles that need to be packed into the minimum number of unit squares. After a long line of works, Bansal and Khan [BK14] gave a 1.405 factor asymptotic approximation algorithm for the problem. On the hardness front, Bansal and Sviridenko [BS04] showed that the problem does not admit an asymptotic PTAS, and this APX hardness result has been generalized to several related problems by Chlebík and Chlebíková [CC06]. We refer the reader to the excellent survey [CKPT17] regarding the geometric problems.

C. Organization

We first define the multidimensional problems and the Label Cover problem formally in Section II. Next, we prove the hardness results for Vector Bin Packing, Vector Scheduling in Section III, Section IV respectively. We defer the Vector Bin Covering part to the full version [San21].

II. PRELIMINARIES

Notations. We use $[n]$ to denote the set $\{1, 2, \dots, n\}$. We use $\mathbf{1}^d$ to denote the d -dimensional vector $(1, 1, \dots, 1)$. For two d -dimensional real vectors $\mathbf{a}$ and $\mathbf{b}$, we say that $\mathbf{a} \geq \mathbf{b}$ if $a_i \geq b_i$ for all $i \in [d]$. For a graph G , we let $\omega(G), \alpha(G), \chi(G)$ be the largest clique size, largest independent size, and the chromatic number of G respectively. A set system or set family $\mathcal{S}$ on a universe V is a collection of subsets of V .

Problem Statements. We give formal definitions of the three problems that we study.

⁶Note that rainbow coloring with 2 colors is the same as proper 2 coloring (Property B) of the hypergraphs.

Definition II.1. (Vector Bin Packing) In the Vector Bin Packing problem, the input is a set of n rational vectors $v_1, v_2, \dots, v_n \in [0, 1]^d$. The objective is to partition $[n]$ into minimum number of parts $A_1, A_2, \dots, A_m$ such that

$$\left\| \sum_{j \in A_i} v_j \right\|_\infty \leq 1 \quad \forall i \in [m]$$

Definition II.2. (Vector Scheduling) In the Vector Scheduling problem, the input is a set of n rational vector jobs $v_1, v_2, \dots, v_n \in [0, 1]^d$, and m identical machines. The objective is to assign the jobs to machines i.e. partition $[n]$ into m parts $A_1, A_2, \dots, A_m$ to minimize the makespan which is defined as the maximum ℓ_∞ load on a machine.

$$\max_{i \in [m]} \left\| \sum_{j \in A_i} v_j \right\|_\infty$$

Definition II.3. (Vector Bin Covering) In the Vector Bin Covering problem, we are given n vectors $v_1, v_2, \dots, v_n \in [0, 1]^d$. The goal is to partition the input vectors into maximum number of parts $A_1, A_2, \dots, A_m$ such that

$$\sum_{j \in A_i} v_j \geq \mathbf{1}^d \quad \forall i \in [m]$$

Asymptotic Approximation. For the Bin Packing problem, it is NP-Hard to identify if all the vectors can be packed into 2 bins or need 3 bins. This already proves that the problem is NP-hard to approximate within $\frac{3}{2}$ as per the usual notion of multiplicative approximation ratio. However, this is less interesting as there are much better asymptotic approximation algorithms for the problem which get $(1 + \epsilon)$ -factor approximation when the optimal value is large enough, for every positive constant $\epsilon > 0$.

Even for the Vector Bin Packing problem, the performance of an algorithm is typically measured in the asymptotic setting. We give the formal definition [CKPT17] of asymptotic approximation ratio of an algorithm $\mathcal{A}$ for the Vector Bin Packing problem.

Definition II.4. (Asymptotic Approximation Ratio) The asymptotic approximation ratio $\rho_{\mathcal{A}}^\infty$ of an algorithm $\mathcal{A}$ for the Vector Bin Packing problem is

$$\rho_{\mathcal{A}}^\infty = \limsup_{n \rightarrow \infty} \rho_{\mathcal{A}}^n, \quad \rho_{\mathcal{A}}^n = \sup_{I \in \mathcal{I}} \left\{ \frac{\mathcal{A}(I)}{\text{OPT}(I)} : \text{OPT}(I) = n \right\}$$

where $\mathcal{I}$ denotes the set of all possible Vector Bin Packing instances.

All the results mentioned in this paper regarding Vector Bin Packing are with respect to the asymptotic approximation ratio.

Label Cover. We define the Label Cover problem:

Definition II.5. (Label Cover) In an instance of the Label Cover problem $G = (V = L \cup R, E, \Sigma_L, \Sigma_R, \Pi)$ with $|\Sigma_L| \geq |\Sigma_R|$, the input is a bipartite graph $L \cup R$ with

constraints on every edge. The constraint on an edge e is a projection $\Pi_e : \Sigma_L \rightarrow \Sigma_R$. We say a labeling $\sigma : V \rightarrow \Sigma_L \cup \Sigma_R$ satisfies the constraint on the edge $e = (u, v)$ if $\Pi_e(\sigma(u)) = \sigma(v)$. The objective is to find a labeling $\sigma : V \rightarrow \Sigma_L \cup \Sigma_R$ that satisfies as many constraints as possible.

By a simple reduction from the 3-SAT problem, we can prove that Label Cover is NP-hard when Σ_L and Σ_R are constants (See e.g., Lemma 4.2 in [BG16]).

Theorem II.6. Given a Label Cover instance when $\Sigma_L = \Sigma_R = [6]$, it is NP-hard to identify if it has a labeling that satisfies all the constraints.

The real use of Label Cover, however, lies in its strong hardness of approximation. PCP Theorem [ALM⁺98] combined with Raz's parallel repetition [Raz98] yields the following strong inapproximability of Label Cover problem.

Theorem II.7. There exists an absolute constant $c > 1$ such that for every integer n and $\epsilon > 0$, there is a reduction from 3-SAT instance I over n variables to Label Cover instance $G = (V = L \cup R, E, \Sigma_L, \Sigma_R, \Pi)$ with $|V| \leq n^{O(\log(\frac{1}{\epsilon}))}$, $|\Sigma_L| \leq (\frac{1}{\epsilon})^c$ satisfying the following:

- 1) (Completeness.) If I is satisfiable, there exists a labeling to G that satisfies all the constraints.
- 2) (Soundness.) If I is not satisfiable, no labeling can satisfy an ϵ fraction of the constraints of G .
- 3) (Biregularity.) The graph $L \cup R, E$ is biregular with degrees on either side bounded by $\text{poly}(\frac{1}{\epsilon})$.

Furthermore, the running time of the reduction is $\text{poly}(n, \frac{1}{\epsilon})$.

Moshkovitz-Raz [MR10] proved the following hardness of near linear size Label Cover.

Theorem II.8. There exist absolute constants $c, c' > 1$ such that for every n and $\epsilon > 0$, there is a reduction from 3-SAT instance I over n variables to Label Cover instance $G = (V = L \cup R, E, \Sigma_L, \Sigma_R, \Pi)$ with $|V| \leq n^{1+o(1)} (\frac{1}{\epsilon})^c$, $|\Sigma_L| \leq 2^{(\frac{1}{\epsilon})^{c'}}$ satisfying the following:

- 1) (Completeness.) If I is satisfiable, there exists a labeling to G that satisfies all the constraints.
- 2) (Soundness.) If I is not satisfiable, no labeling can satisfy an ϵ fraction of the constraints of G .
- 3) (Biregularity.) The graph $L \cup R, E$ is biregular with degrees on either side $\text{poly}(\frac{1}{\epsilon})$.

Furthermore, when ϵ is a constant, the running time of the reduction is $\text{poly}(n)$.

III. VECTOR BIN PACKING

In this section, we prove the hardness of approximation of Vector Bin Packing. First, we define the packing dimension of a set family and bound the packing dimension of simple set families. Next, we combine this upper bound with the

hardness of set cover on simple bounded set systems to prove Theorem 1.1.

A. Packing Dimension

For a set family $\mathcal{S}$ on a universe V , we define the packing dimension $\text{pdim}(\mathcal{S})$ below. For a function $f : V \rightarrow [0, 1]^K$ and a set $S \subseteq V$, we let $f(S)$ denote the vector $f(S) = \sum_{v \in S} f(v)$.

Definition III.1. For a set family $\mathcal{S}$ on a universe V , the packing dimension $\text{pdim}(\mathcal{S})$ is defined as the smallest positive integer K such that there exists an embedding $f : V \rightarrow [0, 1]^K$ that satisfies the following property: For every set $S \subseteq V$, S is in the family $\mathcal{S}$ if and only if

$$\|f(S)\|_\infty \leq 1.$$

If no such embedding exists, we say that $\text{pdim}(\mathcal{S})$ is infinite.

For a set family $\mathcal{S}$ to have finite packing dimension i.e. for an embedding $f : V \rightarrow [0, 1]^K$ realizing the above condition to exist requires two conditions:

- 1) The set family is *downward closed* i.e. for every $S \in \mathcal{S}$ and $T \subseteq S$, $T \in \mathcal{S}$ as well.
- 2) For every element $v \in V$, there is a set $S \in \mathcal{S}$ with $v \in S$. We call a set family $\mathcal{S}$ on a universe V non-trivial if for every $v \in V$, there is a set $S \in \mathcal{S}$ with $v \in S$.

On the other hand, any set system that satisfies the above two conditions i.e. being downward closed and non-trivial has a finite packing dimension. Before proving this statement, we first prove the following simple but useful proposition.

Proposition III.2. For a pair of set families $\mathcal{S}_1$ and $\mathcal{S}_2$ defined on the same universe V such that $\text{pdim}(\mathcal{S}_1)$ and $\text{pdim}(\mathcal{S}_2)$ are finite,

$$\text{pdim}(\mathcal{S}_1 \cap \mathcal{S}_2) \leq \text{pdim}(\mathcal{S}_1) + \text{pdim}(\mathcal{S}_2)$$

Proof: Let $K_1 = \text{pdim}(\mathcal{S}_1)$ and $K_2 = \text{pdim}(\mathcal{S}_2)$. Suppose that $f_1 : V \rightarrow [0, 1]^{K_1}$ be such that for every set $S \subseteq V$,

$$\|f_1(S)\|_\infty \leq 1$$

if and only if $S \in \mathcal{S}_1$. Similarly, let $f_2 : V \rightarrow [0, 1]^{K_2}$ be such that for every set $S \subseteq V$,

$$\|f_2(S)\|_\infty \leq 1$$

if and only if $S \in \mathcal{S}_2$. Consider the function $f : V \rightarrow [0, 1]^{K_1+K_2}$ defined as $f(v) = (f_1(v), f_2(v))$. Then, for every set $S \subseteq V$, $\|f(S)\|_\infty \leq 1$ if and only if $\|f_1(S)\|_\infty \leq 1$ and $\|f_2(S)\|_\infty \leq 1$. Thus, for every set $S \subseteq V$, $\|f(S)\|_\infty \leq 1$ if and only if $S \in \mathcal{S}_1$ and $S \in \mathcal{S}_2$, or equivalently, if $S \in \mathcal{S}_1 \cap \mathcal{S}_2$. Hence, the packing dimension of $\mathcal{S}_1 \cap \mathcal{S}_2$ is at most $K_1 + K_2$. ■

For a set $S \subseteq V$, let $S^\uparrow$ be the family of sets $T \subseteq V$ such that $S \subseteq T$. Similarly, let $S^\downarrow$ be the family of sets $T \subseteq V$

such that $T \subseteq S$. For a set system $\mathcal{S}$, we let $\mathcal{S}^\uparrow$ (resp. $\mathcal{S}^\downarrow$) denote the union of $S^\uparrow$ (resp. $S^\downarrow$) over all $S \in \mathcal{S}$.

Consider a set $S \subseteq V$ with $|S| > 1$. For the set family $2^V \setminus \mathcal{S}^\uparrow$, we have the embedding $f : V \rightarrow [0, 1]$ defined as

$$f(v) = \begin{cases} \frac{1}{|S|} + \frac{1}{|S|^2}, & \text{if } v \in S \\ 0 & \text{otherwise.} \end{cases}$$

This shows that $\text{pdim}(2^V \setminus \mathcal{S}^\uparrow) \leq 1$ for all $S \subseteq V$ with $|S| > 1$. Note that we have

$$\mathcal{S} = \bigcap_{S \notin \mathcal{S}} 2^V \setminus \mathcal{S}^\uparrow$$

for every downward closed set system $\mathcal{S}$. Combined with Proposition III.2, we obtain that for every non-trivial downward closed family $\mathcal{S}$ on a universe V , $\text{pdim}(\mathcal{S}) \leq 2^{|V|}$.

We are interested in the classes of set families for which there is an efficient embedding with packing dimension being independent of $|V|$. In particular, the class of set families that we study are bounded set families where each set has cardinality at most k , and each element appears in at most Δ sets. We can show that such bounded set families that are downward closed and non-trivial have packing dimension at most $(k\Delta)^{O(\Delta)}$. Together with the $\Omega(\log k)$ hardness [Tre01] of k -set cover where each set has cardinality at most k , a fixed constant and each element appearing in $(\log k)^{O(1)}$ sets, this packing dimension bound gives the hardness of $(\log d)^{\Omega(1)}$ for the Vector Bin Packing problem when d is a large constant. Unfortunately, the exponential dependence on Δ is necessary for the packing dimension of bounded set systems, and thus, this approach does not yield the optimal $\Omega(\log d)$ hardness of Vector Bin Packing.

Instead of using arbitrary bounded set families, we bypass this barrier by using simple bounded set families. Recall that a set family is called simple if any two distinct sets in the family intersect in at most one element. It turns out that for simple bounded set families i.e. simple set families $\mathcal{S}$ where each set has cardinality at most k , and each element appears in at most Δ sets, the packing dimension of $\mathcal{S}^\downarrow$ can be upper bounded by $(k\Delta)^{O(1)}$. Together with the $\Omega(\log k)$ hardness of simple k -set cover (proved in the full version [San21]), we get the optimal $\Omega(\log d)$ hardness of Vector Bin Packing when d is a large constant. In the next subsection, we prove the packing dimension upper bound, and we use this upper bound to prove the hardness of Vector Bin Packing in Section III-C.

B. Packing Dimension of Simple Bounded Set Families

The main embedding result that we prove is that the downward closure of simple set systems where each set has cardinality k and each element appears in at most Δ sets has packing dimension at most polynomial in k, Δ .

Theorem III.3. Suppose that $\mathcal{S}$ is a simple non-trivial set system on a universe V where each set has cardinality at most $k \geq 2$ and each element appears in at most Δ sets. Then,

$$\text{pdim}(\mathcal{S}^\downarrow) \leq (k\Delta)^{O(1)}$$

Furthermore, an embedding realizing the above can be found in time polynomial in $|V|$.

We prove the embedding result by writing the set family $\mathcal{S}^\downarrow$ as an intersection of $(k\Delta)^{O(1)}$ structured set families each of which has packing dimension at most $(k\Delta)^{O(1)}$. We can then upper bound the packing dimension of $\mathcal{S}^\downarrow$ using Proposition III.2. The structured set systems we study are *sunflower-bouquets*, which are a disjoint union of sunflowers⁷ that have a single element as the kernel. The formal definition of the sunflower-bouquet set families is below.

Definition III.4. (Sunflower-bouquets) A simple set system $\mathcal{S}$ on a universe V is called a sunflower-bouquet with core $U \subseteq V, U \neq \emptyset$ if the following hold.

- 1) Every set $S \in \mathcal{S}$ satisfies $|S \cap U| = 1$. Furthermore, for every $u \in U$, there is a set $S \in \mathcal{S}$ with $u \in S$.
- 2) For any pair of sets $S_1, S_2 \in \mathcal{S}$ with $S_1 \cap S_2 \neq \emptyset$, we have $S_1 \cap U = S_2 \cap U = S_1 \cap S_2$.

We now give an efficient embedding for a sunflower-bouquet $\mathcal{S}$ on a universe V with core $U \subseteq V, U \neq \emptyset$. The motivation behind this lemma is to upper bound the packing dimension of the set system $\mathcal{T}^\downarrow = \mathcal{S}^\downarrow \cup \{S \subseteq V \setminus U : |S| \leq k\}$. We defer the proof of this lemma to the full version [San21].

Lemma III.5. Fix an integer $k \geq 2$. Let $\mathcal{S}$ be a simple set family defined on a universe V that is a sunflower-bouquet with core U . Furthermore, each set in the family has cardinality at most k and each element appears in at most Δ sets. Then, there exists an embedding $f : V \rightarrow [0, 1]^K$ that satisfies

(A) For every set $S \in \mathcal{S}$,

$$\|f(S)\|_\infty \leq 1.$$

(B) For every set $S \notin \mathcal{S}^\downarrow$ with $S \cap U \neq \emptyset$,

$$\|f(S)\|_\infty > 1.$$

(C) For every set $S \subseteq V$ with $S \cap U = \emptyset$ and $|S| \leq k$,

$$\|f(S)\|_\infty \leq 1.$$

(D) For every set $S \subseteq V$ with $|S| > k$,

$$\|f(S)\|_\infty > 1.$$

⁷A sunflower is a collection of sets $S_1, S_2, \dots, S_r$ whose pairwise intersection is constant i.e., there exists U such that $U = S_i \cap S_j$ for all $i, j \in [r], i \neq j$. This constant intersection U is called the kernel of the sunflower.

with $K = (k\Delta)^{O(1)}$. Furthermore, such an embedding can be found in time polynomial in $|V|$ given $\mathcal{S}$.

As a corollary, we bound the packing dimension of the set family

$$\mathcal{T}^\downarrow = \mathcal{S}^\downarrow \cup \{S \subseteq V \setminus U : |S| \leq k\}.$$

Corollary III.6. Suppose that $\mathcal{T}$ is a set family defined on a universe V with

$$\mathcal{T} = \mathcal{S} \cup \{S \subseteq V \setminus U : |S| \leq k\}$$

where $\mathcal{S} \subseteq 2^V$ is a sunflower-bouquet with core U . Furthermore, each set in $\mathcal{S}$ has cardinality at most $k \geq 2$ and each element appears in at most Δ sets in $\mathcal{S}$. Then,

$$\text{pdim}(\mathcal{T}^\downarrow) \leq (k\Delta)^{O(1)}$$

Furthermore, an embedding realizing this packing dimension can be found in time polynomial in $|V|$ given $\mathcal{S}$.

Proof: As $\mathcal{S}$ is a sunflower-bouquet, from Lemma III.5, there exists an embedding $f : V \rightarrow [0, 1]^K$ that satisfies the conditions (A), (B), (C) and (D) with $K = (k\Delta)^{O(1)}$. Conditions (A) and (C) together imply that

$$\|f(S)\|_\infty \leq 1$$

for all $S \in \mathcal{T}$. Note that

$$\mathcal{T}^\downarrow = \mathcal{S}^\downarrow \cup \{S \subseteq V \setminus U : |S| \leq k\}.$$

Suppose that $S \subseteq V$ is a subset of V with $S \notin \mathcal{T}^\downarrow$. If $S \cap U = \emptyset$, then $|S| > k$, which implies that $\|f(S)\|_\infty > 1$ using condition (D). If $S \cap U \neq \emptyset$, then $S \notin \mathcal{S}^\downarrow$ which implies that $\|f(S)\|_\infty > 1$ using condition (B). Thus, $\|f(S)\|_\infty \leq 1$ if and only if $S \in \mathcal{T}^\downarrow$. ■

We are now ready to prove our main embedding result i.e. Theorem III.3.

Proof of Theorem III.3: We define a graph $G = (V, E)$ as follows: two elements $u, v \in V$ are adjacent in G if there exist sets $S_1, S_2 \in \mathcal{S}$ (not necessarily distinct) such that $u \in S_1, v \in S_2, S_1 \cap S_2 \neq \emptyset$. As the cardinality of each set in $\mathcal{S}$ is at most k and each element of V is present in at most Δ sets, the maximum degree of a vertex in G can be bounded above as

$$\Delta(G) \leq k(k-1)\Delta^2$$

Thus, the chromatic number of G is at most $L = \chi(G) \leq k(k-1)\Delta^2 + 1 \leq k^2\Delta^2$. Using the greedy coloring algorithm, we can partition V into L non-empty parts $U_1, U_2, \dots, U_L$ such that each U_j is an independent set in G . For every $j \in [L]$, as U_j is an independent set in G , we have

- 1) For every set $S \in \mathcal{S}$, $|S \cap U_j| \leq 1$.
- 2) Any two sets $S_1, S_2 \in \mathcal{S}$ with $S_1 \cap U_j \neq \emptyset, S_2 \cap U_j \neq \emptyset$ and $S_1 \cap S_2 \neq \emptyset$ satisfy $S_1 \cap U_j = S_2 \cap U_j = S_1 \cap S_2$.

We now define the set families $\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_L$ as follows:

$$\mathcal{S}_j = \{S \in \mathcal{S} : S \cap U_j \neq \emptyset\} \cup \{S \subseteq V \setminus U_j : |S| \leq k\}$$

We claim that $\bigcap_{j \in [L]} \mathcal{S}_j^\downarrow = \mathcal{S}^\downarrow$. First, consider an arbitrary set $S \in \mathcal{S}^\downarrow$ and an integer $j \in [L]$. As $|S| \leq k$, irrespective of S intersects U_j or not, $S \in \mathcal{S}_j^\downarrow$. Thus, $\mathcal{S}^\downarrow \subseteq \mathcal{S}_j^\downarrow$ for all $j \in [L]$. Consider a non-empty set $S \notin \mathcal{S}^\downarrow$. As $U_1, U_2, \dots, U_L$ is a partition of V , there exists a $j \in [L]$ such that $S \cap U_j \neq \emptyset$. As $S \notin \mathcal{S}^\downarrow$, $S \notin \mathcal{S}_j^\downarrow$. This implies that

$$\bigcap_{j \in [L]} \mathcal{S}_j^\downarrow = \mathcal{S}^\downarrow$$

Using Proposition III.2, in order to bound the packing dimension of $\mathcal{S}^\downarrow$, it suffices to bound the packing dimension of $\mathcal{S}_j^\downarrow$, $j \in [L]$.

Fix an integer $j \in [L]$ and consider the set family $\mathcal{S}_j^\downarrow$. It is defined on the universe V and there exists a non-empty subset $U_j \subseteq V$ such that

$$\mathcal{S}_j = \mathcal{S}'_j \cup \{S \subseteq V \setminus U_j : |S| \leq k\}$$

with

$$\mathcal{S}'_j = \{S \in \mathcal{S} : S \cap U_j \neq \emptyset\}.$$

Here, $\mathcal{S}'_j$ is a simple set system which satisfies the following properties:

- 1) Each set in $\mathcal{S}'_j$ has cardinality at most $k \geq 2$ and each element appears in at most Δ sets in $\mathcal{S}'_j$.
- 2) Every set $S \in \mathcal{S}'_j$ satisfies $|S \cap U_j| = 1$. As $\mathcal{S}$ is non-trivial, for every $u \in U_j$, there exists a set $S \in \mathcal{S}'_j$ with $u \in S$.
- 3) For every pair of sets $S_1, S_2 \in \mathcal{S}'_j$ with $S_1 \cap S_2 \neq \emptyset$, $S_1 \cap U_j = S_2 \cap U_j = S_1 \cap S_2$.

In other words, the set family $\mathcal{S}'_j$ is a sunflower-bouquet with core U_j . Using Corollary III.6, we get that $\text{pdim}(\mathcal{S}'_j) \leq (k\Delta)^{O(1)}$ for all $j \in [L]$, which completes the proof. ■

C. Hardness of Vector Bin Packing

We show that for large enough constant d , Vector Bin Packing is hard to approximate within $\Omega(\log d)$. Our hardness is obtained via the hardness of set cover on simple bounded instances.

In the set cover problem, the input is a set family $\mathcal{S}$ on a universe V with $|V| = n$. The objective is to pick the minimum number of sets $\{S_1, S_2, \dots, S_m\} \subseteq \mathcal{S}$ from the family such that their union is equal to V . The greedy algorithm where we repeatedly pick the set that covers the maximum number of new elements achieves a $\ln n$ approximation factor. Fiege [Fei98] proved a matching hardness of $(1-\epsilon)(\ln n)$. On set systems where each pair of sets intersect in at most one element i.e. simple instances, $\Omega(\log n)$ hardness of set cover is proved by Kumar, Arya, and Ramesh [KAR00]. We observe that by changing the parameters slightly, their reduction also implies the same

hardness on instances where the maximum set size is bounded:

Theorem III.7. (Set Cover on simple bounded instances)

There exists an integer B_0 such that for every constant $B \geq B_0$, the Set Cover problem on simple set systems in which each set has cardinality at most B is NP-hard to approximate within $\Omega(\log B)$. Furthermore, in the hard instances, each element occurs in at most $O(B)$ sets.

The details of the parameter modification appear in the full version [San21].

We combine this set cover hardness with the bound on the packing dimension of simple set systems to prove the hardness of Vector Bin Packing.

Proof of Theorem I.1: We prove the result by giving an approximation preserving reduction from the NP-hard problem of set cover on simple bounded set systems. Let $\mathcal{S}$ be the set system from Theorem III.7 defined on a universe V . Note that each set in the family has cardinality at most $k = B$ and each element in the universe appears in at most $\Delta = O(B)$ sets. We now output a set $\mathcal{V}$ of $|V|$ vectors in $[0, 1]^d$ such that

- 1) (Completeness.) If there is a set cover of size m in $\mathcal{S}$, there is a packing of $\mathcal{V}$ using m bins.
- 2) (Soundness.) If there is no set cover of size m' in $\mathcal{S}$, there is no packing of $\mathcal{V}$ using m' bins.

We use Theorem III.3 to compute an embedding $f : V \rightarrow [0, 1]^d$ in polynomial time such that

$$\|f(S)\|_\infty \leq 1$$

if and only if $S \in \mathcal{S}^\downarrow$, with $d = (k\Delta)^{O(1)} = B^{O(1)}$. Our output Vector Bin Packing instance is the set of vectors $f(v), v \in V$.

$$\mathcal{V} = \{f(v) : v \in V\}$$

Completeness.: Suppose that there exist sets $S_1, S_2, \dots, S_m \in \mathcal{S}$ whose union is V . Then, we use m bins with the vectors $\{f(v_j) : j \in S_i\}$ in the i th bin. A vector might appear in multiple bins, but we can arbitrarily pick one bin for each vector while still maintaining the property that in each bin, the ℓ_∞ norm of the sum of the vectors is at most 1.

Soundness.: Suppose that the minimum set cover in $\mathcal{S}$ has cardinality at least $m' + 1$. Then, we claim that the set of vectors $\mathcal{V}$ needs $m' + 1$ bins to be packed. Suppose for contradiction that there is a vector packing with m' bins. In other words, there exists a partition of V into $B_1, B_2, \dots, B_{m'}$ such that $\|f(B_i)\|_\infty \leq 1$ for all $i \in [m']$. As $\|f(B_i)\|_\infty \leq 1$, $B_i \in \mathcal{S}^\downarrow$ for all $i \in [m']$. That is, for every $i \in [m']$, there exists a set $S_i \in \mathcal{S}$ such that $B_i \subseteq S_i$. This implies that $\{S_1, S_2, \dots, S_{m'}\}$ is a set cover of V , a contradiction.

As the original bounded simple set cover problem is hard to approximate within $\Omega(\log B) = \Omega(\log d)$, the resulting

Vector Bin Packing is hard to approximate within $\Omega(\log d)$. Furthermore, in the hard instances, the optimal value i.e. the minimum number of bins needed to pack the vectors can be made arbitrarily large, and thus, the hardness applies to the asymptotic approximation ratio. ■

IV. VECTOR SCHEDULING

A. Monochromatic Clique

In the Monochromatic Clique problem, given a graph $G = ([n], E)$ and a parameter $k(n)$, the objective is to assign k colors to the vertices of G so as to minimize the largest monochromatic clique. More formally, we study the following decision version of the problem.

Definition IV.1. (Monochromatic-Clique(k, B)) *In the Monochromatic-Clique(k, B) problem, given a graph $G = (V, E)$ with $|V| = n$ and parameters $k(n), B(n)$, the goal is to distinguish between the following:*

- 1) (YES case) *The chromatic number of G is at most k .*
- 2) (NO case) *In any assignment of k colors to the vertices of G , there is a clique of size B , all of whose vertices are assigned the same color.*

It generalizes the standard k -Coloring problem, which corresponds to the case when $B = 2$. Note that the problem gets easier as B increases. Indeed, when $B > \sqrt{n}$, we can solve the problem in polynomial time using the canonical SDP relaxation. We present this algorithm and an almost matching integrality gap in the full version [San21].

On the hardness front, we now prove that Monochromatic-Clique(k, B) is hard when $B = (\log n)^C$, for any constant C . We achieve this in two steps: First, we observe that the existing chromatic number hardness results already imply the hardness of monochromatic clique when $B = (\log n)^\gamma$ for some constant $\gamma > 0$. Next, we amplify this hardness by using lexicographic graph product.

1) *Basic Hardness:* We start with a couple of basic Ramsey theoretic lemmas from [CK04].

Lemma IV.2. *For a graph $G = (V, E)$ with $|V| = n$, if $\omega(G) \leq B$, then $\alpha(G) \geq n^{\frac{1}{B}}$.*

Lemma IV.3. *For a graph $G = (V, E)$ with $|V| = n$, if $\omega(G) \leq B$, then $\chi(G) \leq O(n^{1-\frac{1}{B}} \log n)$.*

We can use the above lemmas to prove that if the chromatic number of a graph is large enough, then in any assignment of k colors to the vertices of the graph, there is a large monochromatic clique.

Lemma IV.4. *For every constant $\epsilon > 0$, if a graph $G = (V, E)$ with $|V| = n$ satisfies $\chi(G) \geq k \frac{n}{2^{(\log n)^\alpha}}$ for some integer k and $0 < \alpha < 1$, then in any assignment of k colors to V , there is a monochromatic clique of size $B = \Omega((\log n)^{1-\alpha-\epsilon})$.*

Proof: Suppose for contradiction that there is an assignment of k colors to V without a monochromatic clique of size B . Using Lemma IV.3, the subgraphs corresponding to each of the k color classes has chromatic number at most

$$O(n^{1-\frac{1}{B}} \log n) = \frac{n}{2^{\Omega((\log n)^{\alpha+\epsilon})}} \log n < \frac{n}{2^{(\log n)^\alpha}}$$

colors. Thus, the whole graph has chromatic number at most $k \frac{n}{2^{(\log n)^\alpha}}$ colors, a contradiction. ■

Khot [Kho01] proved that assuming $\text{NP} \not\subseteq \text{ZPTIME}\left(n^{(\log n)^{O(1)}}\right)$, the chromatic number of graphs is hard to approximate within a factor of $\frac{n}{2^{(\log n)^{1-\gamma}}}$ for an absolute constant $\gamma > 0$. More formally, he proved the following:

Theorem IV.5. ([Kho01]) *There exists a constant $\gamma > 0$, a function $k = k(n)$, and a randomized reduction that takes as input a 3-SAT instance I on n variables and outputs a graph $G = (V, E)$ with $|V| = N = 2^{\log n^{O(1)}}$ such that*

- 1) (Completeness) *If I is satisfiable, $\chi(G) \leq k$.*
- 2) (Soundness) *If I is not satisfiable, with probability at least $\frac{1}{2}$, $\chi(G) > k \frac{N}{2^{(\log N)^{1-\gamma}}}$.*

Furthermore, the reduction runs in time $\text{poly}(N) = 2^{(\log n)^{O(1)}}$.

We observe that Khot's chromatic number hardness immediately gives $(\log n)^{\Omega(1)}$ hardness of Monochromatic Clique.

Lemma IV.6. *There exists a constant $\gamma > 0$, a function $k = k(n)$ such that the following holds. Assuming $\text{NP} \not\subseteq \text{ZPTIME}\left(n^{(\log n)^{O(1)}}\right)$, given a graph $G = ([n], E)$, there is no $n^{(\log n)^{O(1)}}$ time algorithm for Monochromatic-Clique(k, B) when $B = \Omega((\log n)^\gamma)$.*

Proof: Using Khot's reduction, we get that there exists an absolute constant $\gamma > 0$ such that assuming $\text{NP} \not\subseteq \text{ZPTIME}\left(n^{(\log n)^{O(1)}}\right)$, given a graph $G = ([n], E)$ and a parameter $k(n)$, there is no $n^{(\log n)^{O(1)}}$ time algorithm to distinguish between the following:

- 1) (Completeness) $\chi(G) \leq k$.
- 2) (Soundness) $\chi(G) > k \frac{n}{2^{(\log n)^{1-\gamma}}}$.

Using Lemma IV.4, the Soundness condition implies that in any assignment of k colors to G , there is a monochromatic clique of size $\Omega((\log n)^{\gamma-\epsilon})$, for any constant $\epsilon > 0$. Thus, given a graph G and a parameter k , assuming $\text{NP} \not\subseteq \text{ZPTIME}\left(n^{(\log n)^{O(1)}}\right)$, there is no $n^{(\log n)^{O(1)}}$ time algorithm to distinguish between the following:

- 1) (Completeness) $\chi(G) \leq k$.
- 2) (Soundness) In any assignment of k colors to the vertices of G , there is a monochromatic clique of size $\Omega((\log n)^{\gamma'})$.

for any constant $\gamma' < \gamma$. ■

2) *Amplification using Lexicographic Product:* We cannot directly amplify the hardness of the Monochromatic-Clique problem by taking graph products as we cannot preserve the chromatic number and also amplify the largest clique in an assignment of k colors at the same time. We get around this issue by defining a harder variant of Monochromatic Clique called Strong Monochromatic Clique and then amplifying it.

Definition IV.7. (Strong Monochromatic-Clique(k, B, C)) In the Strong Monochromatic-Clique(k, B, C), given a graph G and parameters $k(n), B(n), C$, the goal is to distinguish between the following two cases:

- 1) (YES case) The chromatic number of G is at most k .
- 2) (NO case) In any assignment of k^C colors to the vertices of G , there is a monochromatic clique of size B .

We now observe that the chromatic number hardness of Khot [Kho01] implies the same hardness as Lemma IV.6 for Strong Monochromatic Clique as well.

Lemma IV.8. There exists a constant $\gamma > 0$ and a function $k = k(n)$ such that for every constant $C \geq 1$, the following holds. Assuming $\text{NP} \not\subseteq \text{ZPTIME}(n^{(\log n)^{O(1)}})$, there is no $n^{(\log n)^{O(1)}}$ time algorithm for Strong Monochromatic-Clique(k, B, C) when $B = \Omega((\log n)^\gamma)$.

Proof: Note that the function k in Theorem IV.5 satisfies $k = o(2^{(\log N)^{1-\gamma}})$. Thus, we can replace the soundness condition in Theorem IV.5 with $\chi(G) \geq k^C \frac{N}{2^{C(\log N)^{1-\gamma}}}$. Using Lemma IV.4, this implies that in any assignment of k^C colors to the vertices of G , there is a monochromatic clique of size $\Omega((\log N)^{\gamma-\epsilon})$, where $\epsilon > 0$ is an absolute constant. The hardness of Strong Monochromatic Clique then follows along the same lines as Lemma IV.6. ■

We amplify the hardness of Strong Monochromatic-Clique(k, B, C) to Monochromatic-Clique(k^C, B^C) using the lexicographic product of graphs. First, we define lexicographic product and prove some properties of it.

Definition IV.9. (Lexicographic product of graphs) Given two graphs G and H , the Lexicographic graph product $G \cdot H$ has vertex set $V(G) \times V(H)$, and two vertices $(u_1, v_1), (u_2, v_2)$ are adjacent if either $(u_1, u_2) \in E(G)$ or $u_1 = u_2$ and $(v_1, v_2) \in E(H)$.

The lexicographic product can be visualized as replacing each vertex of G with a copy of H and forming complete bipartite graphs between copies of vertices adjacent in G . For ease of notation, we let $G^2 = G \cdot G$. More generally, for an integer n that is a power of 2, we define G^n as taking the above lexicographic product of G with itself recursively $\log n$ times.

Lemma IV.10. Let $n \geq 2$ be a power of 2. If $\chi(G) \leq k$, then $\chi(G^n) \leq k^n$.

Proof: We prove that $\chi(G^2) \leq k^2$, and the statement follows by induction on n . If $f : G \rightarrow [k]$ is a proper k -coloring of G , then the coloring $f'(u, v) = (f(u), f(v))$ is a proper k^2 -coloring of $G \times G$. ■

Lemma IV.11. Let $n \geq 2$ be a power of 2. Suppose that in any assignment of k colors to the vertices of G , there is a monochromatic clique of size B . Then, in any assignment of k colors to the vertices of G^n , there is a monochromatic clique of size B^n .

Proof: We prove the statement for $n = 2$ and the lemma follows by induction on n . Let $f : V(G^2) \rightarrow [k]$ be a given assignment. For a vertex $v \in G$, consider the assignment $g_v : V(G) \rightarrow [k]$ defined as $g_v(u) = f(v, u)$. As every assignment of k colors to the vertices of G has a monochromatic clique of size B , there is a color $\alpha(v) \in [k]$ and a clique $S(v) \subseteq V(G)$ with $|S(v)| \geq B$ such that $g_v(u) = \alpha(v)$ for all $u \in S(v)$, or in other words, $f(v, u) = \alpha(v)$ for all $u \in S(v)$. Note that such a set $S(v)$ and $\alpha(v)$ exist for $v \in V(G)$. The function $\alpha : V(G) \rightarrow [k]$ can also be visualized as an assignment of k colors to the vertices of G , and thus there is a monochromatic clique T of size at least B with respect to this assignment. The set

$$\{S(v) : v \in T\}$$

is a monochromatic clique of size B^2 with respect to f in G . ■

By using the lexicographic product, we can get a polynomial time reduction from Strong Monochromatic Clique to Monochromatic Clique.

Lemma IV.12. For every constant $C \geq 1$ that is a power of 2, there exists a polynomial time reduction from Strong Monochromatic-Clique(k, B, C) to Monochromatic-Clique(k^C, B^C).

Proof: Given a graph G as an instance of Strong Monochromatic-Clique(k, B, C), we compute the graph $G' = G^C$. We claim that solving Monochromatic-Clique(k^C, B^C) on G' solves the original Strong Monochromatic Clique problem.

- 1) (Completeness.) Suppose that $\chi(G) \leq k$. Then, by Lemma IV.10, $\chi(G') \leq k^C$.
- 2) (Soundness.) Suppose that in any assignment of k^C colors to the vertices of G , there is a monochromatic clique of size B . Then, by Lemma IV.11, in any assignment of k^C colors to the vertices of G' , there is a monochromatic clique of size B^C . ■

Putting everything together, we obtain the following hardness of Monochromatic Clique.

Theorem IV.13. For every constant $C > 0$, there exists a function $k = k(n)$ such that the following holds. Assuming $\text{NP} \not\subseteq \text{ZPTIME}\left(n^{(\log n)^{O(1)}}\right)$, there is no $n^{(\log n)^{O(1)}}$ time algorithm for Monochromatic-Clique(k, B) when $B = \Omega((\log n)^C)$.

Proof: The proof follows directly by combining Lemma IV.8 and Lemma IV.12. ■

B. From Monochromatic Clique to Vector Scheduling

We now prove Theorem I.2 using the above hardness of Monochromatic Clique.

Proof of Theorem I.2: The reduction from Monochromatic-Clique(k, B) to Vector Scheduling is (implicitly) proved in [CK04]. We present it here for the sake of completeness. Given a graph $G = (V = [n], E)$, parameters k and B , we order all the B -sized cliques of G as $T_1, T_2, \dots, T_d$ with $d \leq n^B$. We define a set of n vectors $v_1, v_2, \dots, v_n$ of dimension d with

$$(v_i)_j = \begin{cases} 1 & \text{if } i \in T_j \\ 0 & \text{otherwise.} \end{cases}$$

The instance of the Vector Scheduling has these n vectors as the input and the number of machines is equal to k .

We analyze the reduction.

- 1) (Completeness.) Suppose that there exists a proper k -coloring of G , $c : V \rightarrow [k]$. We assign the vector v_i to the machine $c(i)$. For every $j \in [d]$, all the B vectors that have 1 in the j th dimension are assigned to distinct machines. Thus, the makespan of the scheduling is at most 1.
- 2) (Soundness.) Suppose that in any assignment of k colors to the vertices of G , there is a monochromatic clique of size B . In this case, the makespan of the scheduling is at least B .

We set $B = (\log n)^C$ for a large constant C to be set later. We choose k from Theorem IV.13 such that assuming $\text{NP} \not\subseteq \text{ZPTIME}\left(n^{(\log n)^{O(1)}}\right)$, there is no $n^{(\log n)^{O(1)}}$ time algorithm for Monochromatic-Clique(k, B). By the above reduction, we can conclude that there is no polynomial time algorithm that approximates the resulting Vector Scheduling instances within a factor of $B = (\log n)^C$. As $d \leq n^B$, we get that $\log d \leq (\log n)^{C+1}$, and $B \geq (\log d)^{1-\frac{1}{C+1}}$. Setting $C = \frac{1}{\epsilon} - 1$, we get that d -dimensional Vector Scheduling has no polynomial time $\Omega((\log d)^{1-\epsilon})$ approximation algorithm assuming $\text{NP} \not\subseteq \text{ZPTIME}\left(n^{(\log n)^{O(1)}}\right)$, for every constant $\epsilon > 0$. ■

Remark IV.14. In [IKKP19], Im, Kell, Kulkarni, and Panigrahi also study the ℓ_r -norm minimization of Vector Scheduling where the objective is to minimize

$$\max_{k \in [d]} \left(\sum_{i=1}^m (L_i^k)^r \right)^{\frac{1}{r}}$$

where L_i^k denotes the load on the machine i on the k th dimension. They gave an algorithm with an approximation ratio $O\left(\left(\frac{\log d}{\log \log d}\right)^{1-\frac{1}{r}}\right)$. Our reduction from Monochromatic Clique gives almost optimal hardness for this variant as well: we get the hardness of $\Omega\left((\log d)^{1-\frac{1}{r}-\epsilon}\right)$ assuming $\text{NP} \not\subseteq \text{ZPTIME}\left(n^{(\log n)^{O(1)}}\right)$, for every constant $\epsilon > 0$.

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